Choupo

The Open Source Chemical Process Simulator

Properties Guide

Choupo-dev

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"Thermodynamics is a funny subject. The first time you go through it, you don't understand it at all. The second time you go through it, you think you understand it, except for one or two points. The third time you go through it, you know you don't understand it, but by that time you are so used to it, it doesn't bother you any more."

— Arnold Sommerfeld

1 Introduction: the difficulty, the map, the promise

What this guide is — and is not. This is the props-**mode workflow** guide: how you *estimate* a missing compound, *fit* its binary pairs, and *consistency-test* the data — the see-then-decide loop of the Props/Explore workspace. It is *not* the property *theory*. For the first-principles derivations of NRTL, Wilson, UNIQUAC, UNIFAC, the cubic equations of state (SRK, Peng–Robinson), the electrolyte models (Pitzer, eNRTL, aqueous speciation), Henry's law, Rackett, Ambrose–Walton, sublimation and the NASA–CEA Gibbs-of-formation data, see the **Theory Guide** (the in-GUI model panels deep-link straight into it).

If thermodynamics felt slippery the first time — the formalism pristine, yet somehow never self-sufficient when you sat down with real data — you are in good company: Sommerfeld, who reshaped quantum physics, said he never wrote his thermodynamics book because he never felt he had understood the subject. The difficulty is real, but it is not formal incompleteness. It is that an *axiomatic-looking* macroscopic theory rests, underneath, on emergent statistical facts and on parameters that must be *measured*, not derived. The confusion is the seam between the two showing through. This guide does not hide that seam — it makes it the organising idea.

1.1 The target: where theory grips

Picture property prediction as a target (Figure 1): pristine at the core, increasingly empirical toward the rim.

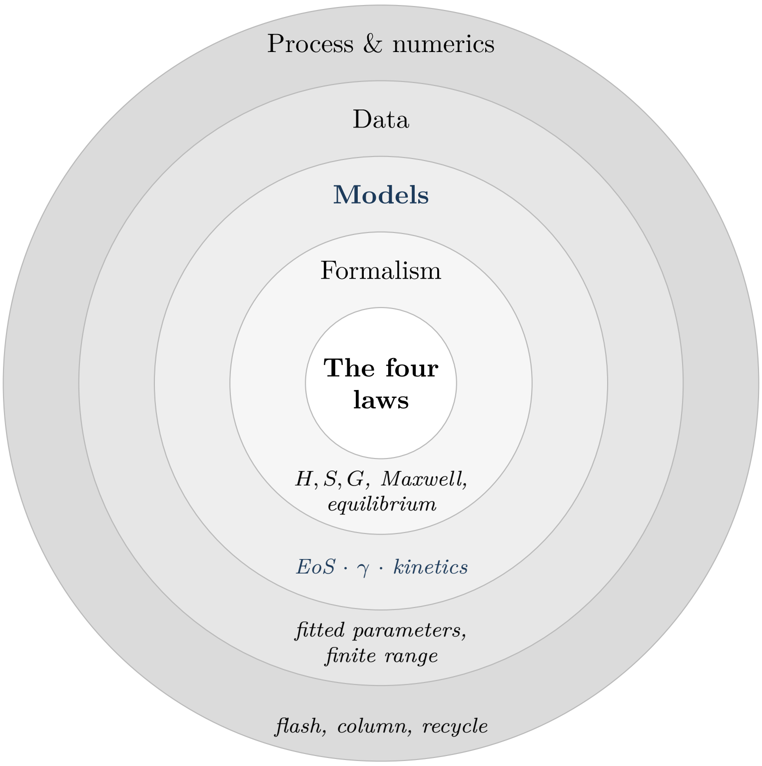

Figure 1: The layers of property prediction. The two inner shells follow from the laws alone. The accent shell — *constitutive models* — is the first fault line: the laws say *that* fugacities must match across phases, never *how* to compute one for a real mixture. From there outward, the theory is necessary but no longer sufficient.

- **The four laws (core).** The zeroth, first, second and third laws define temperature, internal energy, entropy, and — via Nernst — an absolute zero of entropy. Parameter-free.
- **Formalism.** Pure manipulation yields the potentials H, G, A, the Maxwell relations, the equilibrium criteria (equality of fugacity across phases, $\Delta G = -RT \ln K$ for reaction) and the phase rule. Still derivable from nothing but the laws.
- **Constitutive models — where it first grips.** The laws require fugacities to be equal across phases; they do not tell you how to *compute* a fugacity for a real mixture. You must supply a model they do not give you — an equation of state, an activity model, a kinetic law. In Choupo: `idealGas/SRK/PR`; `ideal/NRTL/Wilson/UNIFAC`.
- **Data — "go to the laboratory."** Those models carry parameters thermodynamics cannot predict: T_c, P_c, ω, the Antoine constants, $C_p^\circ(T)$, ΔH_f°, S°, the NRTL a_{ij}. Each is measured or estimated, and each has a finite

validity range. In Choupo, these are the `.dat` files.

- **Process & numerics.** Where the predicted numbers are consumed — flash, distillation, recycle — and grip a second time, now numerically (multiple EoS roots, phase stability, initialisation).

> **Key idea.** Sommerfeld's difficulty is this picture: a theory that *looks* axiomatic to the rim, but stops being self-sufficient at the accent shell. The honest move is not to paper over the seam between rigorous derivation and fitted correlation — it is to show exactly where it runs.

1.2 Where the models came from — a genealogy

The chain in §3 shows there are exactly *two* ways to a fugacity in a mixture: through an equation of state (φ) or through an activity coefficient (γ). Each has its own lineage, and knowing them explains why the model list looks the way it does — and why it is a *list* at all rather than a single answer.

The equation-of-state line (1873 →). Van der Waals wrote one equation for both phases, continuous through the critical point: the first time liquid and gas were the same substance in one formula. Redlich–Kwong (1949) improved the attraction term; Soave (1972) made it obey the vapour-pressure curve by introducing the acentric factor; Peng–Robinson (1976) repaired the liquid density. All four share one skeleton — repulsion plus attraction, with the parameters generated from T_c, P_c, ω — which is why adding a new cubic to Choupo touches *no data file*. Then the molecular branch: Wertheim's association theory (1984) led to SAFT (1989) and PC-SAFT (2001), where a molecule is a *chain of segments* and the parameters are $m, \sigma, \varepsilon/k$ instead of the critical triad. That is exactly why PC-SAFT extrapolates in temperature and density where a cubic drifts — and why it cannot borrow the cubic's parameters.

The activity-coefficient line (1964 →). Regular-solution theory failed on polar mixtures. Wilson (1964) introduced *local composition* — the admission that a molecule does not see its neighbours in bulk proportions — but could not describe a liquid–liquid split. NRTL (Renon & Prausnitz, 1968) added non-randomness and *could*: that is why an LLE case in this guide reaches for NRTL and not Wilson. UNIQUAC (Abrams & Prausnitz, 1975) split the excess Gibbs energy into a combinatorial part (size and shape) and a residual part (energy). UNIFAC (Fredenslund, 1975) made UNIQUAC *predictive* by decomposing molecules into functional groups, so a pair never measured could still be estimated. COSMO-RS (Klamt, 1995) and COSMO-SAC (Lin & Sandler, 2002) went further and replaced the group table with the molecule's own screening-charge surface from a quantum calculation — the first genuinely *a priori* route to γ, and the one that needs no binary datum at all.

The electrolyte line (1923 →). Debye–Hückel derived the ionic atmosphere and is *exact* in the dilute limit — and only there. Davies (1962) stretched it empirically to a few tenths molal. Pitzer (1973) abandoned the pretence of derivation and wrote a virial expansion in molality that holds to seawater and beyond. Edwards *et al.* (1978) extended that expansion to *molecular* solutes — dissolved NH_3, CO_2, H_2S — which a charge-only correlation cannot price at all, since $z = 0$ makes it return $\gamma = 1$ identically. eNRTL (Chen, 1982) carried local composition into ionic solutions, which is what makes a mixed solvent — salt plus water plus ethanol — tractable at all.

Choosing among the three aqueous models. The bench offers `davies`, `pitzerHMW` and `edwardsPitzer` as the `activityModel` of a `speciate` operation, and they are not interchangeable:

- `davies` — one line, no parameters, trustworthy to about $I \approx 0.5$ mol/kg and *announced* as indicative beyond it. Every neutral solute gets exactly 1. Use it for a first look and for teaching what a dilute correlation is.
- `pitzerHMW` — the Harvie–Møller–Weare parameterisation, per ion, with ternary mixing. The right tool for brines, scaling and seawater.
- `edwardsPitzer` — a *different* truncation of Pitzer's expansion, extended to molecules, and the one its $NH_3/CO_2/H_2S/SO_2$ parameters were regressed in. Use it for weak-electrolyte and sour-water systems.

The last point generalises and is worth stating once: **a parameter set belongs to the equation it was fitted in.** Pitzer's expansion admits several truncations, and a β regressed under one of them is not a β for another. Quoting a published number inside a different model is a category error that no amount of agreement in one test case repairs, which is why Choupo carries the two Pitzer engines side by side under distinct names rather than merging their parameter tables.

> **Key idea.** Each model was invented because the previous one *failed* at something specific. The list is not a menu of equivalent options; it is a history of failures, each one repaired. Choosing a property method is therefore choosing *which failure you can live with* in your system — which is why this architecture insists that the choice be declared in the case file, verified at assembly, and confessed in the log. A simulator that

hides the choice behind a dropdown has not removed the failure; it has removed your view of it.

1.3 The glass-box promise

Because the outer shells are empirical, every number Choupo predicts should travel with two things attached:

- **Provenance** — which method produced it (rigorous derivation, fitted correlation, or group-contribution estimate) and the source of each parameter (a measured datum or an estimate).
- **Validity awareness** — whether a correlation is *extrapolating beyond its fitted range*, with a warning when it is, rather than a confident number where none is warranted.

> *Intuition.* Ask Choupo for the vapour pressure of nitrogen at room temperature and it *refuses*: N_2 is supercritical there (its critical temperature is −147 °C), and a saturation pressure does not exist above the critical point. The P^{sat} curve simply *breaks* at T_c instead of extrapolating Antoine into a meaningless ∼600 bar. That refusal is the whole promise in miniature: the model declines to invent a number the physics forbids.

1.4 Where we are heading

A single property over a wide temperature span is not one problem — it is several physical regimes stitched together, and your confidence in any predicted number rises and collapses as you cross them. Figure 2 is the destination of this guide: the vapour pressure of a **water / ethanol / benzene** mixture from −200 to 1000 °C, drawn not as a curve but as a band of *confidence*.

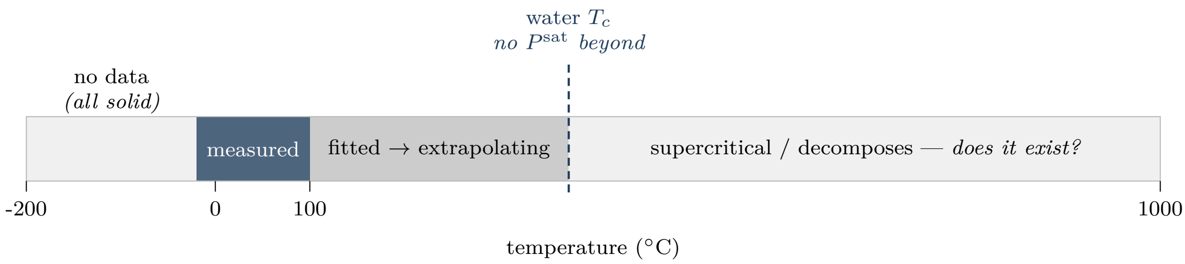

Figure 2: The same property, five honesties. A narrow window where we actually have data; degradation outward as the correlation extrapolates; a hard *wall* at the critical temperature where the property ceases to exist (the nitrogen refusal, generalised); and a cryogenic desert where every species has frozen solid (ethanol last, at −114 °C) and there is simply no data. The theory runs to both edges; reality and data do not.

We build each piece of that picture in the chapters that follow — estimating a missing component (§6), curating the mixture models against lab points (§7), and testing whether the data itself is even consistent (§8). We read that span regime by regime at the end of this introduction (§1.8), and return to solve it numerically, end to end, at the close.

1.5 Anatomy of a component: where the numbers come from

Open any `.dat` and you see a flat list of numbers. The natural question — the one this whole guide turns on — is: *which of these are independent inputs, and which derived properties does the engine actually build from them?* Let us read `acetone.dat` field by field and follow each number to the property it feeds and to the line of code that consumes it.

Field	**What it is, and what it builds**
Identity	
`MW`	molar mass; converts molar ↔ mass everywhere.
Caloric anchors — `standardThermochemistry`	
`dHf_298` ΔH_f°	the *position* of the enthalpy scale: bond energy C_p knows nothing about (combustion calorimetry).
`s_298` S°	the *absolute* (third-law) entropy at 298 K — a true zero-referenced value, not a difference.
Caloric shape — `idealGasHeatCapacity`	
`coefficients` $C_p^\circ(T)$	the master correlation: integrated to give the *temperature shape* of H, S, G (below).
Corresponding-states trio	
`Tc`, `Pc`	critical point: set the *size* of the cubic-EoS attraction term a_c, and the reduced temperature $T_r = T/T_c$.
`omega` ω	acentric factor: sets the *temperature shape* of that attraction, $\alpha(T_r;\omega)$ — how non-spherical / polar the molecule is.
Phase change	
`Tb`, `HvapTb`	boiling point + latent heat at T_b; with `Tc` drive Watson's $\Delta H_{\mathrm{vap}}(T)$.
`Vliq`	liquid molar volume (Poynting, liquid density).
Correlations & recipes	
`vaporPressure`	Antoine $\log_{10} P^{\mathrm{sat}} = A - B/(T + C)$, with its `Trange`; the saturation curve.
`groups`	the molecular decomposition a group-contribution estimator / UNIFAC reads (the curation recipe, §6).
`diffusionVolume`	Fuller gas diffusivity ⇒ Lewis number (wet-bulb).

The caloric engine: H, S, G from C_p

This is the mechanism behind "other properties that derive from here" — and it is worth seeing *how one arrives at* $S(T)$, since that is the step that trips everyone up. Start from the laws, at constant pressure. Heat capacity *is* the temperature slope of enthalpy, $C_p = (\partial H/\partial T)_P$, so

$$dH = C_p\, dT.$$

For entropy, the second law gives $dS = \delta q_{\mathrm{rev}}/T$; at constant pressure the reversible heat is $\delta q_{\mathrm{rev}} = C_p\, dT$, hence

$$dS = \frac{C_p}{T}\, dT.$$

That single step — dividing by T — is the whole origin of the $\int C_p/T$. Integrate each from a reference temperature and add the 298 K anchor; Choupo stores $C_p^\circ(T)$ and the two anchors (*not* tables of H, S) and evaluates:

$$H^\circ(T) = \underbrace{\Delta H_f^\circ}_{\text{position}} + \int_{298.15}^{T} C_p^\circ(T')\, dT' \qquad (\texttt{Component::h\_pure\_ig},\ \texttt{Component.cpp}) \tag{1}$$

$$S^\circ(T) = \underbrace{S^\circ_{298}}_{\text{absolute}} + \int_{298.15}^{T} \frac{C_p^\circ(T')}{T'}\, dT' \qquad (\texttt{Component::s\_pure\_ig}) \tag{2}$$

$$G^\circ(T) = H^\circ(T) - T\, S^\circ(T) \tag{3}$$

For a mixture the engine adds the entropy of mixing $-R\sum y_i \ln y_i$ and the pressure term $-R\ln(P/P_{\mathrm{ref}})$ (`ThermoPackage::S_ig`). That is the *entire* ideal-gas caloric model — no hidden tables.

These integrals are not done numerically. For a polynomial C_p the primitive is elementary — it was solved *once, on paper* — so each call evaluates a closed form,

$$\int_{T_{ref}}^{T} \frac{C_p^\circ}{T'}\, dT' = a_0 \ln\frac{T}{T_{ref}} + \sum_{k\geq 1} \frac{a_k}{k}\left(T^k - T_{ref}^k\right),$$

exact to machine precision in a handful of operations (`PolynomialCp::S`). The standard correlations all share this property (NASA, Aly–Lee, Shomate have closed primitives too); Choupo never runs a quadrature loop for C_p.

> **Worked example 1.1.** Acetone H and S at 400 K — by hand, then by ChoupoWith $C_p^\circ(T) = 6.301 + 0.2606\,T - 1.253\times10^{-4}T^2 + 2.038\times10^{-8}T^3$ (units J mol^{-1}K^{-1}), integrating from 298.15 to 400 K:
>
> $$\int C_p^\circ\, dT' = 8431\ \text{J/mol}, \qquad \int \frac{C_p^\circ}{T'}\, dT' = 24.19\ \text{J/(mol}\cdot\text{K)}.$$

So $H^\circ(400) = -217100 + 8431 = \mathbf{-208669}$ J/mol and $S^\circ(400) = 295.30 + 24.19 = \mathbf{319.49}$ J/(mol · K). Running `choupoProps` on a one-point scan returns exactly $H_{\text{ig}} = -208669$ J/mol, $S_{\text{ig}} = 319.49$ J/(mol · K). The number on the page is the number in the code — and at 298.15 K both integrals vanish, so $H = \Delta H_f^\circ$ and $S = S^\circ_{298}$, the anchors laid bare.

Key idea. C_p gives the *shape*; the formation values give the *position*. That is why the `.dat` lists $C_p^\circ(T)$ *and* ΔH_f°, S° separately — not redundancy. And note the asymmetry: entropy has an absolute zero for free (third law: $S \to 0$ for a perfect crystal at $T = 0$), so S°_{298} is a genuine absolute; enthalpy has *no* natural zero, so ΔH_f° must be *measured* and supplied.

What the critical properties and acentric factor actually do

T_c, P_c, ω do nothing in the caloric equations above — they enter only when you ask for *real-fluid* behaviour through a cubic equation of state. There the attraction term is $a(T) = a_c\,\alpha(T_r)$, with a_c fixed by T_c, P_c and the temperature shape fixed by ω (Soave/SRK):

$$\alpha(T) = \left[\,1 + m(1 - \sqrt{T_r})\,\right]^2, \quad m = 0.48508 + 1.55171\,\omega - 0.15613\,\omega^2, \quad T_r = \tfrac{T}{T_c}$$

(`SRK.cpp`; Peng–Robinson uses the analogous $\kappa(\omega)$). This $a(T)$ is what produces the departure functions $H - H^\circ$, $S - S^\circ$ and the compressibility Z — the real-gas correction layered on top of the ideal-gas caloric result. The same T_c, T_b also drive Watson's latent heat, $\Delta H_{\text{vap}}(T) = \Delta H_{\text{vap}}(T_b)\,[(T_c - T)/(T_c - T_b)]^{0.38}$ (`Component.cpp`).

> *Intuition.* So why does `acetone.dat` carry ω at all if you are running the default `idealGas`? Because the data is *model-agnostic* and the model decides which fields wake up. Under `idealGas` there is no departure, and ω, T_c, P_c are simply never read. Select `SRK` for a high-pressure compressor and the very same numbers come alive. The `.dat` is the laboratory record; the model is the question you ask of it.

The entropy ledger, completed and verified

The real-fluid entropy the isentropic machines consume is the ideal-gas assembly above plus the model's departure,

$$S(T, P, y) \;=\; S^{\text{ig}}(T, P, y) \;+\; S^{\text{R}}(T, P, y),$$

with S^{R} supplied by whichever equation of state the case declares: SRK carries the closed form $S^{\text{R}} = R\ln(Z - B) + (\mathrm{d}a/\mathrm{d}T/b)\,\ln\!\big((Z + B)/Z\big)$ (Sandler's eq. 6.4-31, cited in `SRK.cpp`), Peng–Robinson its generalised-cubic analogue, and IF97 water evaluates its own fundamental-equation entropy directly. One convention divergence is on record rather than hidden: PC-SAFT's `S_residual` returns the constant-density (T, ρ) residual *without* the $+R\ln Z$ conversion to the (T, P) departure the cubics use — stated in its own source comment, flagged in `docs/design/entropy-glass-box-trace.md`, and open until a witnessed fix; no shipped tutorial prices an isentropic machine on PC-SAFT today. The phase-crossing legs of the pure-component chain — and why the vaporisation step at the 298 K datum is *not* $\Delta H_{\text{vap}}/T$ — are derived in the Theory Guide's chapter on integral properties.

The whole chain is pinned by a witness the reader can run: `tutorials/props/molecular/entropy01_air_ledger` publishes every line for air at 400 K and 2 bar — $s_i^\circ(T)$ per component, S^{ig}, S^{R}, S — and its golden holds the re-addition pure + mixing + pressure to the engine's own total. The "What is entropy?" EduTool page performs that re-addition on screen from a live run of the same case.

And the ledger buys a price tag: the `exergy` bench operation evaluates the physical exergy $b = (h - h_0) - T_0\,(s - s_0)$ from the same H and S surfaces at a *declared* dead state (`deadState { T0; P0; }` — the environment is the case author's fact, and a missing block refuses rather than assuming one). Because only differences at fixed composition enter, the enthalpy datum, the s°_{298} anchors, the mixing term and the reference pressure all cancel: b is datum-independent. Witness `exergy01_air_dead_state` prices the same air state at 2136.7 J/mol and pins the structural zero ($b = 0$ at the dead state itself, on both legs). Chemical exergy is deliberately refused by name: it needs a standard-environment model (Szargut) that is a curated data decision, not a formula.

1.6 The ledger, explained: `explainProperty`

The witness above publishes the lines; `explainProperty` publishes where each line *came from*. For one property at one `state` (`S_real`, `S_ig`, `H_real`, `H_ig`) it lists every term with the record field or engine call behind it — the s°_{298} or ΔH_f° datum, the $\int C_p/T$ term as the *difference* of two engine calls, the mixing and pressure lines, the EoS residual — re-adds them, and checks the sum against the engine. It computes no thermodynamics of its own; a gap beyond round-off refuses. In `entropy01_air_ledger` the `explainS` op re-adds air's S_{ig} at 400 K and 2 bar, $203.068 + 4.273 - 5.763 = 201.578$ J/(mol K), `gap_total` pinned at 0; the `explainH` twin has no mixing and no pressure line. Refused by name: an unknown property, a datum off the ideal-gas rung, and a state answered by a `pureFluids {}` override, whose ledger is the release's own equation.

1.7 From G to equilibrium: what ΔG° is (and is not)

The single most common confusion in this whole subject is ΔG° versus ΔG. It is worth settling once, because G is what the reactors minimise and it is built from exactly the H and S above.

First, the $^\circ$. It means *reference state* — each species pure, ideal gas, at 1 bar — but *at the reaction temperature* T. It does *not* mean "at 298 K." So $\Delta G^\circ = \Delta G^\circ(T)$ varies with temperature; $\Delta G^\circ(298)$ is just one point on a curve.

Then, the two are not the same thing.

- ΔG° is the Gibbs change if reactants → products with *everything in its standard state*. It is a *fixed number* — a property of the reaction at T.
- ΔG is the *actual* Gibbs change at the real composition (the real partial pressures). It is the *driving force*.

The reaction quotient Q bridges them, and equilibrium is where the driving force — not the reference number — vanishes:

$$\Delta G = \Delta G^\circ + RT\ln Q, \qquad \Delta G = 0 \text{ at equilibrium } \Rightarrow \boxed{\Delta G^\circ = -RT\ln K}$$

> **Key idea.** At equilibrium it is ΔG that is zero, *never* ΔG°. ΔG° stays fixed and merely sets *where* the equilibrium sits (the value of K); the system gets there by shifting Q until $\Delta G = 0$. And $\Delta G^\circ > 0$ does not mean "no reaction" — it means $K < 1$, i.e. partial conversion. Only the sign and size of K change.

Where the number comes from — straight back to the caloric engine:

$$\Delta G^\circ(T) = \Delta H^\circ(T) - T\,\Delta S^\circ(T) = \sum_i \nu_i\, g_i^\circ(T), \qquad g_i^\circ(T) = h_i^{\mathrm{ig}}(T) - T\, s_i^{\mathrm{ig}}(T).$$

Note the subtlety that confuses everyone: h^{ig} carries the *formation* enthalpy (ΔH_f°, an elements datum) while s^{ig} is the *absolute* (third-law) entropy. Mixing the two references looks wrong — but it is exactly right, because in a *balanced* reaction the element contributions cancel through $\sum_i \nu_i$. That is precisely why the `.dat` carries ΔH_f° *and* S° separately: each plays a different role, and they combine correctly only across a balanced stoichiometry (`Component::g_pure_ig`; consumed in `PFR.cpp`/`CSTR.cpp` as $K = \exp(-\Delta G^\circ/RT)$ and in the Gibbs reactor's direct G-minimisation).

How K moves with T (Gibbs–Helmholtz). $\Delta G^\circ(T)$ is not a frozen datum — it has its own temperature law, and that law is a workhorse relation worth knowing by name. Gibbs–Helmholtz states $[\partial(G/T)/\partial T]_P = -H/T^2$; applied to $\Delta G^\circ = -RT\ln K$ it collapses to the van 't Hoff equation,

$$\boxed{\frac{d\ln K}{dT} = \frac{\Delta H^\circ}{RT^2}}$$

which is just Le Chatelier made quantitative: an exothermic reaction ($\Delta H^\circ < 0$) has K *falling* as T rises. In Choupo it doubles as a glass-box self-check — the $K(T)$ the reactors build from $g^\circ(T)$ at each temperature must show exactly this slope, or the enthalpy and the equilibrium constant are disagreeing with each other.

> *Intuition.* These relations are not ornaments. Gibbs–Helmholtz is *how the equilibrium moves with temperature* (above); its sibling, the **Gibbs–Duhem** equation $\sum_i x_i\,\mathrm{d}\ln\gamma_i = 0$ (constant T, P), says the activity coefficients of a mixture are *not independent*. That single constraint is exactly why measured VLE data can be *tested* for thermodynamic consistency — and why an activity model is built from one G^{E} function rather than fitting each γ_i freely. It is the engine of §8.

1.8 A worked span: water / ethanol / benzene, -200 to 1000 °C

Put the three together and sweep the temperature. This one mixture crosses *every* regime of property prediction, and is the honest way to see both how properties are obtained *and* where the models give out — which is exactly the colour band of Figure 2, now read regime by regime. (Critical temperatures: ethanol 241 °C, benzene 289 °C, water 374 °C; melting points: ethanol −114, water 0, benzene 5.5 °C.)

Regime	**How a property is obtained**	**Where the model gives out**
-200 to $\sim 0\,°C$ solids (SLE)	melting points T_m, ΔH_{fus}, solid C_p; freezing-point depression; sublimation P^{sat}	**low-T edge.** Below every Antoine `Trange`: P^{sat} would extrapolate blindly. Solid-phase data is sparse — a cryogenic *data desert.*
~ 0 to $\sim 65\,°C$ two liquids (LLE)	activity model (`NRTL`/`UNIFAC`) for γ; water/benzene nearly immiscible, ethanol the cosolvent	VLE-fitted parameters do *not* serve LLE; `Wilson` cannot represent a liquid split at all; without T-dependent parameters the miscibility gap appears/vanishes wrongly.
~ 65 to $\sim 100\,°C$ boiling (VLLE)	Antoine P^{sat} + γ–ϕ + three-phase flash (Gibbs minimisation); ethanol/water + ternary azeotrope ($\sim 64.9\,°C$)	azeotrope sensitivity — a small γ error moves the column design; bubble-point (Wang–Henke) distillation is unstable *through* an azeotrope (use `model simultaneous`).
~ 100 to $\sim 290\,°C$ vapour $\to$ near-critical	cubic EoS (`SRK`/`PR`) with k_{ij}; departure functions $H - H°$, $S - S°$, Z	cubics degrade as the critical region is approached; k_{ij} often needs to be made temperature-dependent over so wide a span.
~ 290 to $374\,°C$ mixed super/sub-critical	the method itself must switch γ–$\phi \to \phi$–ϕ (one EoS for every phase)	above a species' T_c the activity-coefficient picture loses meaning; no single γ model spans the switch; near-critical accuracy is poor.
374 to $1000\,°C$ supercritical + reaction	ϕ–ϕ EoS; wide-range $C_p^\circ(T)$ for the caloric terms	**high-T edge.** P^{sat} ceases to exist above T_c (Choupo *refuses* it, §1.2); and the species thermally *decompose* — the fixed-component assumption fails: do they still exist?

The two edges — the ones you asked to watch — fail for opposite reasons but teach the same lesson. At the **cold edge** there is simply no data: every correlation is fitted to a window that starts well above 73 K, the species are frozen, and a confident number there would be invention, not prediction. At the **hot edge** the theory keeps producing numbers — Antoine will gladly report a vapour pressure at 800 °C — but the *physics* has moved on: past each T_c no saturation exists, and past a few hundred degrees the molecules are no longer the molecules in the `.dat`. In both cases the correct output is a *flag*, not a figure.

Common pitfall. No single model spans this axis. A run that quietly uses one γ model and one cubic from -200 to $1000\,°C$ is not "general" — it is extrapolating through three regime changes in silence. The wide span is not one problem; it is five problems wearing one temperature axis.

> *Intuition.* What Choupo ships *today* covers the middle of this span honestly: cubic EoS, `ideal`/`NRTL`/`Wilson`/`UNIFAC` activity, Antoine P^{sat}, Watson ΔH_{vap}, three-phase VLLE by Gibbs minimisation, and azeotropic columns via the simultaneous MESH solver. Since 2026-08-03 it also carries PC-SAFT with Wertheim association (the 2B and 4C schemes; Theory Guide, the PC-SAFT chapter) for the hydrogen-bonding non-ideality, and the electrolyte engines — Pitzer, eNRTL, the Harvie–Møller–Weare and Edwards truncations — for the ionic edge. What it does *not* carry is a wide-range near-critical package, and no association or electrolyte model here is a general-purpose black box: each is declared by the case, announced by the run, and parameterised from a cited record. So for the near-critical edge, and for any mixture the curated parameters do not reach, Choupo's job is not to fake accuracy it does not have — it is to predict where it can and to *say so* where it cannot. That honesty is the product.

1.9 The low-T edge on a real case: `freezingPoint`

The solid row above, on a case. `freezingPoint` traces the freezing-point curve $T_f(m)$ of a single salt in water as chemical-potential equality between ice and the solution's water,

$$f_{\text{solid}}(T) \;=\; a_w(m,T)\,P^{\text{sat}}(T),$$

the left side an actual crystallising `SolidPhase`, the right the Pitzer salt surface's water activity. Routing through the phase makes its refusals (no ΔH_{fus}, no triple point) and its announced approximations (the neglected heat-capacity term, the sub-freezing P^{sat} extrapolation) fire on a real run. It publishes the curve, T_f and the depression at the grid end, the dilute-limit slope against νK_f (water's *derived* cryoscopic constant) and the AAD against a `validation` dataset. A grid point whose bracket holds no root is refused by name — a eutectic the single-salt model does not contain. Witness `fpd01_nacl_freezing`: NaCl to 2.0 mol/kg, T_f = 266.325 K there (depression 6.835 K), slope/$\nu K_f = 0.931$, one interim measured point (AAD 0.046 %).

2 The Choupo property architecture — the complete map

Everything in this guide hangs off one architecture. This chapter is the complete map, stated once; every later chapter is a room in this house.

2.1 The flow

> `constant/thermoPhysPropDict` *declares* → the builder *loads, verifies and announces* (it never estimates) → the engine *computes* → the run log *confesses*.

Estimation is a curation-time act (the props bench writes a reviewable `.dat`); assembly-time refusal is loud (a missing declared parameter file, a k_{ij} regressed for another cubic, a vapour method the liquid path cannot serve). The log then shows the whole assembly: the VLE world, the reference rung of every group, and every pair parameter with its source file.

2.2 The grammar — the axes of a property package

Axis	**Slot**	**What it f**
components	`components ( ... )`	identity; a ions
phases	`phases`; the component's own `solidPhases{}` block	which phas `phases/sc`
model per PHASE	`equilibrium { formulation ...; liquid {...} vapour {...} }`	ONE Gib `formulati`
convention per GROUP	reference rungs (`referenceBasis`)	renormalis Raoult, so infinite dil
correlation per COMPONENT	the component `.dat` (+ overrides)	pure-speci pure correl surface
parameter per PAIR	`parameters { ...Pairs {...} }`	declared F to its sourc
mixing per PROPERTY × PHASE	`mixing` slots + `effective{}`	how pure Nissan, var closures (a viscosity) l are *media* ing
chemistry	`chemistry { salts (...); }`	real equilil
package per UNIT	unit `thermo {}`	the spatial whole pacl single axes in one pha dited (ΔH

phase equilibrium	`phaseEquilibrium {}`	for each
		imposed
		DATUM
		`datum aqu`
		`<pair-fil`
		surfaces,
		implicit

Provenance is an *attribute of every value*, not an axis: each resolved number carries a quality word (`measured > fitted > correspondingStates > estimated > asserted`), and an `asserted` value — a pinned constant, a user polynomial, a user table — is refused without its written `why`.

2.3 The five formulations

A case declares one `equilibrium.formulation`; the builder assembles that one natively and gives every other shape a named refusal. Each is validated against measured data in this guide:

Formulation	Liquid method	K-value	Natura
gammaPhi	activity.NRTL (Wilson, UNIQUAC, UNIFAC, cosmoSAC)	$\gamma_i P_i^{sat}/P$	organic
gammaGamma	one activity model *per named liquid phase*	$\gamma_i^\alpha/\gamma_i^\beta$	LLE / V
diluteSolution	solution.henryDilute	$\gamma_i^* H_i/P$	dissolved
phiPhi	eos.SRK / eos.PengRobinson / PCSAFT	φ_i^L/φ_i^V	high pre
electrolyteGammaPhi	Pitzer / eNRTL (molality)	ion activities, osmotic	brines, s

`gammaGamma` is the one that surprises people: two coexisting liquids are *two phases*, each entitled to its own Gibbs surface, and the split is found by minimising G directly rather than by iterating K — a symmetric γ-model would otherwise collapse onto the trivial $K = 1$ saddle. That is the legality rule of §3 doing visible work.

2.4 The five component routes

1. **Standard, by name** — `data/standards/components/` (exact-name $O(1)$, every value cited, the engine refuses to write there).
2. **Local, by name** — `data/local/components/` is the *private* working tier (gitignored; shipped empty). Records are runtime-usable, but every use is announced as `[local] UNVERIFIED`; a local record never shadows a standard one. Integrity and provenance gates make a local record inspectable, not automatically authoritative.
3. **New, defined in the case** — `myCase/constant/components/`: a new molecule (estimated and labelled), a new ion, a pseudo-component lump. The case runs self-contained from day one; promotion into the standards tree is a *requested, reviewed curation act* after validation.
4. **Standard + overlay** — a partial case file merges field-by-field over the standard entry (sample-specific measured data, a new model block); the run announces the merge.
5. **Solids** — solid reference rungs, solids on their own stream channel, and the iron rule: a dissolving salt's solid formation enthalpy is ION-DERIVED ($H_f^{solid} = \sum \nu_i h_{f,aq,i} - \Delta H_{soln}$), never a stored duplicate.

Thus component resolution is **case-local overlay > standard > local (private, announced)**. This is an evidence ladder, not merely a search path: moving a file up the ladder requires review and validation, not a rename.

2.5 The ChemSep import route

The ChemSep 8.3 importer (`bin/curate/chemsep_to_choupo.py`) is deliberately reproducible and reviewable. It runs CAS-based de-duplication against the standard catalogue, then a structural/provenance audit and an internal $P^{\mathrm{sat}}(T_b)$ closure check on every imported component record — a closure check on the imported constants and predictive correlation, not an independent comparison with laboratory data.

The same import stages NRTL, UNIQUAC and Wilson binary-pair records, each passing identity, direction, licence, source-digest and exact unit-conversion checks. Their source does not provide a captured temperature-validity range, so they remain screening data and the runtime prints `[local] UNVERIFIED` when one is selected. Pair resolution is **case-local/inline > standard > local > announced ideal default**; local data can therefore extend a case without silently changing an approved standard.

The source boundary is intentional. The unmodified third-party database and its Artistic-2.0 licence are kept under `thirdParty/chemsep/` (gitignored); the normalised, attributed records are written to the *private* `data/local/` tier — they are never redistributed with the public repository. Literature values retain literature provenance; Ambrose–Walton and other closures retain predictive provenance. Missing C_p, formation, electrolyte, adsorption, kinetic or transport data are not inferred merely to make a record look complete.

The case additionally owns its *equipment physics*: reaction kinetics in `constant/reactions` (the heat of reaction always computed from the formation datum and announced), per-machine measured curves (a dryer's falling-rate data) in the unit's own folder, all inherited through the LOUD fractal cascade (plant → sector → unit).

2.6 The reference case

The whole architecture is exercised by one designed flagship, `tutorials/plant/lithiumBrinePlant` (under construction): salar brine through Pitzer ponds and crystallisers, NF membrane Mg/Li separation, solvent extraction with a student-defined extractant (the γ–φ world and the audited Pitzer↔NRTL seam), electrodialysis, 60-bar CO_2 carbonation (the φ–φ override and the Henry world), a radical-kinetics burner satellite for thermal NO_x, batch recrystallisation recipes, a control satellite, and the full outer ring (fitting, sweeps, optimisation, sizing, costing, economics). Later chapters refer to its sectors by name.

2.7 The principle

No silent crutch, anywhere: every value is cited, estimated-and-labelled, overlaid-and-announced, or asserted-with-a-why — and the run log is where the whole assembly confesses. There is no ultimate model; hunting with the cat is honest engineering, hiding the cat is fraud.

3 The thermodynamic chain: how every equation here is derived

Every K-value, activity coefficient and Henry constant in this guide falls out of ONE chain of ideas. This section walks it once, so that no equation below ever has to be taken on faith. (Full derivations: the Theory Guide, chapters on flash, activity models, electrolytes and Henry's law.)

3.1 1. Equilibrium is a Gibbs minimum

The second law says an isolated system maximises entropy; recast at the conditions a process engineer actually controls — fixed T and P — it says a system minimises its **Gibbs energy** G. For a multicomponent, multiphase system,

$$dG \;=\; -S\,dT + V\,dP + \sum_i \mu_i\,dn_i, \qquad \mu_i \equiv \left(\frac{\partial G}{\partial n_i}\right)_{T,P,n_{j\neq i}},$$

and at the minimum, moving a mole of i between phases must change nothing:

$\boxed{\mu_i^\alpha = \mu_i^\beta}$ for every component, across every pair of phases.

That single boxed statement is ALL the physics; everything else is bookkeeping to make it computable.

3.2 2. Fugacity: the workable currency

μ_i is awkward: its absolute value is unknowable and it dives to $-\infty$ as $x_i \to 0$. Lewis (1901) defined the **fugacity** f_i through $\mu_i = \mu_i^\circ + RT\ln(f_i/f_i^\circ)$ — an "effective pressure" that equals y_iP in an ideal gas. Because ln is monotonic, the equilibrium condition becomes

$$\boxed{f_i^\alpha = f_i^\beta}$$

— equal fugacities, phase by phase, component by component. Every VLE "world" in Choupo is just a different, self-consistent way of WRITING f_i for a phase.

3.3 3. Two routes to a mixture fugacity

Route A — an equation of state (works for any phase the EoS can represent). Define the fugacity coefficient $\varphi_i \equiv f_i/(y_iP)$; thermodynamics gives it exactly from the volumetric behaviour,

$$RT\ln\varphi_i \;=\; \int_V^\infty \left[\left(\frac{\partial P}{\partial n_i}\right)_{T,V,n_j} - \frac{RT}{V}\right] dV \;-\; RT\ln Z,$$

so ONE model of $P(T,V,\mathbf{n})$ — a cubic like SRK with its $a(T)$, b and the pair correction k_{ij} — yields φ_i for BOTH the vapour root and the liquid root of the same cubic.

Route B — an activity model (liquids only). Split the liquid fugacity into a reference times a correction, $f_i^L = \gamma_i\,x_i\,f_i^\circ$, where the **activity coefficient** γ_i comes from a model of the excess Gibbs energy (G^E: NRTL, UNIQUAC, Wilson, Pitzer, eNRTL) via $RT\ln\gamma_i = \partial G^E/\partial n_i$. The price of Route B is choosing the **reference** f_i° — and that choice is a CONVENTION, not physics:

Convention	**Reference** f_i°	**Normalised so**
symmetric (Raoult)	pure liquid: $P_i^{\rm sat}$ ($\times\,\varphi_i^{sat}$·Poynting at high P)	$\gamma_i \to 1$ as $x_i \to 1$
unsymmetric (Henry)	infinite dilution: $H_{i,\rm solv} = \lim_{x\to 0} f_i/x_i$	$\gamma_i^* \to 1$ as $x_i \to 0$
electrolyte (aqueous)	ion at infinite dilution, $H^+(aq) = 0$ datum	$\gamma_\pm \to 1$ as $m \to 0$

A supercritical solute HAS no pure liquid, so its only definable anchor is the Henry limit — which is why dissolved H_2/N_2 take the unsymmetric convention, never an Antoine curve extrapolated above T_c.

3.4 4. One Gibbs surface per phase — the legality rule

Gibbs–Duhem, $\sum_i x_i\,d\ln\gamma_i = 0$ at constant T, P, ties every component's γ in a phase to every other's: they must all derive from ONE G^E (or one EoS) surface. Hence the rule that governs Choupo's `thermoPhysPropDict` grammar: *models per phase; conventions per component-group within the phase; correlations per component; parameters per pair.* Mixing two γ-models (or two cubics) in one phase violates Gibbs–Duhem and is refused. Mixing CONVENTIONS in one phase is legal — solvent symmetric, solutes unsymmetric — because a convention only renormalises the same surface.

3.5 5. The four worlds' K-values, assembled

Setting $f_i^L = f_i^V$ with each route/convention gives every K-value this guide uses:

world	f_i^L	f_i^V	$K_i = y_i/x_i$
γ-φ	$\gamma_i x_i P_i^{\text{sat}}$	$y_i \varphi_i P$	$\dfrac{\gamma_i\, P_i^{\text{sat}}}{\varphi_i\, P}$
Henry (KK/KI)	$\gamma_i^* x_i H_i(T)\, e^{\bar{v}_i^\infty (P-P_s)/RT}$	$y_i \varphi_i P$	$\dfrac{\gamma_i^* H_i\, \mathcal{P}}{\varphi_i P}$
φ-φ	$x_i \varphi_i^L P$	$y_i \varphi_i^V P$	$\dfrac{\varphi_i^L}{\varphi_i^V}$
electrolyte	ion activities from Pitzer/eNRTL on the aqueous rung; full speciation (mass action + charge balance) where the case demands it		

Each row is one `equilibrium.formulation` (`gammaPhi`, `diluteSolution`, `phiPhi`, `electrolyteGammaPhi`); the run log announces which world is live. All four are validated in the tutorials against measured data — van Ness-consistent VLE sets (γ-φ), Wiebe/Larson solubilities (Henry, $\sim$2%), the Stryjek N_2–CH_4 isotherm (φ-φ, $\sim$2%), and the Hamer–Wu / calorimetric electrolyte sets.

> *Intuition.* The whole edifice is: G minimum $\to$ equal μ $\to$ equal f $\to$ pick HOW to write f per phase (an EoS or a G^E model + a reference convention) $\to$ a K-value. When an equation in this guide looks arbitrary, walk it back up this chain — it never is.

3.6 Route A, decomposed: `mixingRules`

A cubic prices a mixture through two numbers, and the one-fluid rule builds them from n pures and $n(n-1)/2$ binary corrections:

$$a_{\text{mix}}(T) = \sum_i \sum_j y_i\, y_j\, (1 - k_{ij})\sqrt{a_i a_j}, \qquad b_{\text{mix}} = \sum_i y_i\, b_i.$$

`mixingRules` publishes that decomposition at one `state`: every pure $a_i(T)$ and b_i, every k_{ij} as constructed, every pair's term, and the identity that re-adds them to the a_{mix} the engine runs. The numbers come from `EquationOfState::mixingLedger`, a window on the *same* `buildMix` the hot path runs — the op transcribes no rule of its own, and a re-addition gap beyond round-off refuses. Witness `mixrules01_natural_gas` (CH_4/N_2/CO_2, 250 K): one declared k_{ij} (N_2–CH_4, 0.0289) beside two pairs on the announced $k_{ij} = 0$ default; a_{mix} and b_{mix} identical at 30 and 1 bar while Z moves from 0.9967 to 0.9010. Refused by name: no equation-of-state slot, or a model with no one-fluid rule — ideal gas has no a, PC-SAFT mixes inside its dispersion sum.

3.7 The K-values, traced: `phaseEnvelope`

`phaseEnvelope` draws the P–T envelope of a mixture at fixed feed `composition`: the bubble curve ($V/F \to 0$) and the dew curve ($V/F \to 1$), marched in `pressure`, written as CSV. The saturation solves are `BubblePoint::compute` and `DewPoint::compute`, the `bubbleT`/`dewT` units' own routines. The cricondentherm and the retrograde region live at the nose, exactly where a single-specification march dies, so the op does *not* pretend to close the nose: each branch is traced while its specification stays single-valued, and the last point is a stopping point, never an extremum. An *interior* temperature maximum is reported as a *bracketed* cricondentherm; a converged point discontinuous with its branch (a root jump) ends the branch. Witnesses: `envelope01_natural_gas` (40 dew points, cricondentherm bracketed at 274.705 K, bubble branch never started) and `envelope02_root_jump` (bubble stopped at 18 bar, dew at 26 bar). Michelsen's arc-length continuation is the named, unimplemented next step.

4 Why a separate guide

The Theory Guide answers *how a model works* — how a flash splits, how an EoS predicts Z, how Levenberg–Marquardt minimises a residual. This guide answers a question that comes *before* that: *where do the numbers that feed the models come from?* A simulation is only as good as the T_c, the NRTL pair, the heat of formation it stands on, and those are not handed down — they are estimated, fitted, and tested.

Choupo treats that as a first-class phase with its own binary (`choupoProps`) and its own lifecycle:

> **CREATE** a component (from groups) → **CURATE** the mixture models (fit pairs, test data, choose by evidence) → **WIRE** the flowsheet → **SIMULATE**.

A `propsDict` of `operations` drives the first two stages. A case may have *no* `flowsheetDict` at all (a pure properties case — the compound bench), or carry one and use props to curate the thermo it will simulate. Same files, same engine; only the centre of the GUI differs.

> *Intuition.* The usual reflex is to click a property method and trust a badge. Choupo's stance (the "see, then decide" principle) is the opposite: you make the estimate or the fit *visible* — the group build-up, the model curve against the lab points, the consistency residual — and decide from the evidence in front of you. Estimation has error; the point is to see it.

5 Where curated data lives, and how a case selects it

You estimated a T_c, fitted an NRTL pair, tested a heat of solution. Two questions follow: *where does that number land*, and *how does a case use it*? Choupo's answer is *declarative property packages with OpenFOAM-like explicit files.*

A curated value lands in a glass-box `.dat` in a tree organised by *what kind of thing* it is — the intrinsic property of one substance, the parameter of a pair, the equilibrium of a reaction:

```
data/standards/
  components/<name>.dat            intrinsic, one substance (Tc, Antoine, Cp)
                                   + dissociatesTo { Na 1; Cl 1; }  [salts]
  species/<name>.dat               one modelSpecies file per species
                                   (Na, Cl, CO2aq, O2): formula + charge +
                                   medium-tagged thermo; never fed to a flowsheet
  (a crystal phase lives in its COMPONENT's own solidPhases{} block:
   rho_p, k_v and the dissolution equilibrium of its own solid --
   the phases/solid/ directory is RETIRED, legacy read only)
  chemistry/<name>.dat             FLAT; an equilibrium: dissolution,
                                   speciation (recordType names the family)
  parameters/.../<i>-<j>.dat       a pair: NRTL, Pitzer, eNRTL
  (methods/ is RETIRED: the reference basis is a consequence of the
   declared formulation -- the one-knob rule)
  (the system is declared INLINE in the case's constant/thermoPhysPropDict
   -- THE CENTRE; there is no shared package catalogue)
```

The arity rule decides *which* file: a number that "stands alone with the molecule" (the T_c of water) lives with the component; a number that needs two names (an NRTL pair) cannot, and goes to `parameters/`. The Developer Guide (*the thermophysical property architecture*) gives the full layout and the eight data kinds.

Declaring the system. A case does not list loose models and parameters. It *declares* its thermophysical system — one record that names the components, the formulation, and the parameter files the run needs (active chemistry lives beside it in `constant/chemistryDict`):

```
recordType    thermophysicalPropertySystem;
schemaVersion 2;

components ( water NaCl );          // NaCl's dissociatesTo names the ions

equilibrium
{
    formulation electrolyteGammaPhi;

    aqueous
    {
        solvent            water;
        apparentComponents ( NaCl );  // stream carries the salt; the model
        activityModel { model Pitzer; }        //   activates its ions
        compositionBasis molality;
    }

    vapour { fugacityModel idealGas; }
}
% Pitzer pairs resolve by (cation,anion) name from parameters/; declared
% binaryParameters/binaryInteractions files load from their ‘source‘ paths
% and refuse when missing.
```

The flow is one sentence: **`thermoPhysPropDict` declares, `ThermoPackageBuilder` assembles, `ThermoPackage` computes.** The declaration is the recipe; the builder loads what it names and mounts the runtime object the unit operations consume. An ordinary molecular case is the same record degenerate — no dissociation, no chemistry, a `gammaPhi` block with an NRTL or an EoS slot.

Intuition. The boundary you must respect: *curation estimates, runtime does not.* You may estimate, fit, and derive at the props bench — and you *see* it (§6, §7). Once a case runs, the builder only *loads* existing case, standard or local records; it never invents a missing property. For activity models, an unresolved binary pair may use the documented ideal default, but that fallback is announced and appears in the provenance. A required record with no defined fallback stops the run with a clear message. So: estimate, look, promote — and inspect every local or defaulted rung used by the package.

6 Creating a component: Joback group contribution

When a component is missing from the catalogue and from the literature, you can *estimate* its pure-component constants from its molecular structure — the groups it is built from. Choupo implements the **Joback** method (Joback & Reid, 1987; tabulated in Poling, Prausnitz & O'Connell, *The Properties of Gases and Liquids*, 5th ed.), the `estimateComponent` property operation.

6.1 The method

Each first-order group contributes a tabulated increment. Summing the increments and substituting into Joback's correlations gives the constants (with n_A the total number of atoms):

$$T_b = 198.2 + \sum_i n_i \, \Delta T_{b,i} \tag{4}$$
$$T_c = T_b \Big/ \big(0.584 + 0.965\,\Sigma\Delta T_c - (\Sigma\Delta T_c)^2\big) \tag{5}$$
$$P_c = \big(0.113 + 0.0032\,n_A - \Sigma\Delta P_c\big)^{-2} \quad \text{[bar]} \tag{6}$$
$$V_c = 17.5 + \sum_i n_i \, \Delta V_{c,i} \quad [\text{cm}^3/\text{mol}] \tag{7}$$
$$\Delta H_f^\circ(298) = 68.29 + \sum_i n_i \, \Delta H_{f,i} \quad \text{[kJ/mol, ideal gas]} \tag{8}$$
$$\Delta H_{\text{vap}}(T_b) = 15.30 + \sum_i n_i \, \Delta H_{v,i} \tag{9}$$
$$C_p^{\text{ig}}(T) = (\Sigma a - 37.93) + (\Sigma b + 0.210)T + (\Sigma c - 3.91\times 10^{-4})T^2 + (\Sigma d + 2.06\times 10^{-7})T^3. \tag{10}$$

The acentric factor ω is *not* a group sum; Choupo derives it from (T_b, T_c, P_c) by the Lee–Kesler correlation, with $T_{br} = T_b/T_c$ and P_c in atm:

$$\omega = \frac{-\ln P_c - 5.97214 + 6.09648/T_{br} + 1.28862\ln T_{br} - 0.169347\,T_{br}^6}{15.2518 - 15.6875/T_{br} - 13.4721\ln T_{br} + 0.43577\,T_{br}^6}. \tag{11}$$

A few first-order group increments (Joback & Reid; the full table is in the code, `src/propertyOps/EstimateComponent.cpp`):

group	ΔT_b	ΔT_c	ΔP_c	ΔH_f (kJ/mol)
$-\mathrm{CH_3}$	23.58	0.0141	−0.0012	−76.45
$>\mathrm{CH_2}$	22.88	0.0189	0.0000	−20.64
−OH (alcohol)	92.88	0.0741	0.0112	−208.04
>C=O (ketone)	76.75	0.0380	0.0031	−133.22
aromatic =CH−	26.73	0.0082	0.0011	2.09

6.2 The structure is identity: the `groups{}` block and `bin/estimate`

The molecular structure belongs *with the substance*, like the formula: a component's `.dat` declares its groups once, method-keyed inside one `groups{}` block,

```
groups
{
    joback ( { group CH3; count 2; } { group ketone; count 1; } );
    unifac ( { group CH3; count 1; } { group CH3CO; count 1; } );
}
```

(one home for every decomposition — Joback and UNIFAC are different tables, so each method keys its own list), and estimation needs no case at all:

```
bin/estimate acetone          # case-local first, then the standard catalogue
bin/estimate path/to/new.dat  # or an explicit file
```

The command runs the Joback engine on the component's own groups, prints the full glass-box build-up (each group's increment, the closure chain, deviations against any `reference` block), and drops *one* stable proposal — `<name>.estimated.dat` — next to the source, for you to review and promote by hand. It never writes into the frozen `data/standards/` tree, and the solver still refuses to estimate at run time: estimation is a *curation* act with a paper trail, not a silent fill-in.

The `estimateComponent` operation remains the pedagogical form — declare the groups inline in a `propsDict` when you want the estimate itself to be the lesson, as in the worked example below.

6.3 Worked example: acetone

Acetone, $CH_3-CO-CH_3$, decomposes into $2\times$ `CH3` and $1\times$ `ketone`. The operation prints the build-up and validates against a `reference` block:

Property	Joback	Reference (NIST)	Dev.
T_b	322.1 K	329.2 K	−2.2%
T_c	500.6 K	508.1 K	−1.5%
P_c	48.0 bar	47.0 bar	+2.2%
V_c	209.5 cm^3/mol	209 cm^3/mol	≈ 0
ω	0.295	0.307	−3.9%
ΔH_f°	−217.8 kJ/mol	−217.1 kJ/mol	+0.3%

Within a few percent on the criticals and almost exact on ΔH_f — and you *see* the error, because the reference sits beside the estimate.

> *Intuition.* **Know the method's limits.** Joback is weak on T_b of strongly hydrogen-bonding species (ethanol's T_b comes out $\sim$14 K low), and it yields *no* Antoine / vapour-pressure parameters — those need a fit (§7) or a corresponding-states correlation, a separate step. An estimate is a starting point you then anchor to data, never a final answer.

> **Runnable case.** `tutorials/props/estimate/estimate_acetone` — the example above, groups in, validated estimate out.

7 Curating mixture models: fitting binary pairs

A pure-component set is not enough for a mixture: the non-ideality lives in the *binary interactions* (NRTL, Wilson). Those parameters are regressed against experimental VLE data by Levenberg–Marquardt — the `fitParameters` operation.

7.1 The activity-coefficient models

For a binary, the **NRTL** model gives the liquid-phase activity coefficients from three adjustable numbers per pair — τ_{12}, τ_{21} (temperature-dependent) and the non-randomness α:

$$\tau_{ij} = a_{ij} + \frac{b_{ij}}{T}, \qquad G_{ij} = \exp(-\alpha\,\tau_{ij}), \tag{12}$$

$$\ln\gamma_1 = x_2^2\left[\tau_{21}\left(\frac{G_{21}}{x_1 + x_2 G_{21}}\right)^2 + \frac{\tau_{12}\,G_{12}}{(x_2 + x_1 G_{12})^2}\right], \tag{13}$$

and $\ln\gamma_2$ by swapping the indices. So a pair carries four fittable coefficients $(a_{12}, b_{12}, a_{21}, b_{21})$ plus a fixed α (often 0.3). **Wilson** is the two-parameter alternative,

$$\ln\gamma_1 = -\ln(x_1 + \Lambda_{12}x_2) + x_2\left[\frac{\Lambda_{12}}{x_1 + \Lambda_{12}x_2} - \frac{\Lambda_{21}}{x_2 + \Lambda_{21}x_1}\right], \tag{14}$$

which cannot represent a liquid–liquid split (use NRTL when one is suspected).

7.2 Predicting without pairs: COSMO-SAC

NRTL and Wilson need a fitted pair — somebody measured that binary. **COSMO-SAC** is the predictive alternative: the activity coefficients come from each compound's *sigma profile* — the histogram of screening charge density over its molecular surface, from a quantum-chemical calculation done *once, elsewhere, at curation time* — and no binary parameter exists at all. Choupo implements the 2002 variant (Lin & Sandler), following the NIST benchmark implementation, as a plain activity model: same interface as NRTL, selected the same way.

A compound carries its profile in its own `.dat`, as one or more *named parameter sets* (a compound may carry profiles from several public collections, each labelled with its source):

```
cosmo
{
    VT2005            // set name = its source collection
    {
        variant       cosmoSAC2002;
        source        "Mullins et al., IECR 45 (2006) 4389";
        area          [-] 106.3;      // A^2
        volume        [-] 118.8;      // A^3
        sigmaProfile  ( ... );        // 51 points, -0.025..0.025 e/A^2
    }
}
```

The case selects the model and, when a compound carries more than one set, names which one:

```
activityModel { model cosmoSAC;  source VT2005; }
```

A compound with a single set needs no `source` (its lone set is the default); a multi-set compound without one is a loud error, as is a missing `cosmo` block or a set whose declared `variant` does not match the implemented constants — a profile is never silently combined with another variant's parameters. The standard catalogue ships VT2005 sets for 77 components. Validation: a pure component gives $\ln\gamma = 0$ exactly, and the implementation is cross-checked bit-for-bit against an independent code on the same equations and profiles. What Choupo deliberately does *not* do is generate profiles (that is quantum chemistry, a curation act outside the simulator) or bulk-import thousands of compounds. Reference case: `tutorials/props/molecular/cosmoSAC01_water_ethanol`.

7.3 The residual the fit minimises

The measurable is the bubble temperature. At a fixed pressure P and liquid composition x, the bubble point is the T that closes modified Raoult,

$$\sum_i x_i\,\gamma_i(x,T)\,P_i^{\text{sat}}(T) = P \tag{15}$$

(a Newton-1D in T; P_i^{sat} from Antoine). Levenberg–Marquardt then drives the pair coefficients $\theta = (a_{12}, b_{12}, a_{21}, b_{21})$ to minimise the sum of squared bubble-temperature residuals against the data,

$$\min_{\theta} \sum_{k} \left[T_{\text{bub}}(x_k; \theta) - T_k^{\text{exp}} \right]^2. \tag{16}$$

> *Intuition.* **A low residual is not a good fit.** At a *single* pressure $\tau_{ij} = a_{ij} + b_{ij}/T$ is nearly linear in both a_{ij} and b_{ij}/T over the narrow T range of one isotherm/isobar, so a and b are *correlated* — many (a, b) pairs fit equally well and the optimum is a valley, not a point. `fitParameters` forms $J^{\mathsf{T}} J$ at the optimum and reports the parameter correlation + confidence intervals so you SEE this; the Theory Guide's *Levenberg–Marquardt* and *What is fittable* chapters derive the diagnostic. Fit across pressures, or fix b, to break it.

7.4 See, then decide: the model overlay

Before fitting, look. Declare one scan `operation` per candidate model (ideal, NRTL, Wilson), emitting the $(x, T_{\text{bubble}}, y_{\text{eq}})$ trio, and an `experimental {}` entry that overlays them — and, with a `dataset`, draws the measured points on top. The student sees which model the data actually picks: for ethanol/water at 1 atm the ideal curve misses the azeotrope entirely while NRTL lands on it.

> **Runnable case.** `tutorials/props/compare/compare_vle_etoh_water` — ideal vs. NRTL bubble/dew curves overlaid with measured VLE points (Carey & Lewis, 1932).

7.5 Fit, then promote

When a model earns its keep, `fitParameters` regresses its coefficients to the data and writes a proposal to `constant/parameters/<model>/<pair>.fit-<date>.dat`. Promotion is a deliberate, visible act: you inspect the parity plot and the identifiability statistics (confidence intervals, parameter correlation) the fit reports — never a "good fit" badge — and copy the proposal into the case's `constant/` when you trust it.

8 Testing the data itself: consistency

Before fitting a model *to* data, test whether the data is even thermodynamically consistent — a fit to bad data is a polished wrong answer. The `vleConsistency` operation does this in three steps.

8.1 Step 1 — activity coefficients straight from the data

It does not assume a model. From each measured (x_1, y_1, T) at the run pressure P, modified Raoult gives the activity coefficient of each component directly:

$$\gamma_i = \frac{y_i\,P}{x_i\,P_i^{\mathrm{sat}}(T)}, \qquad P_i^{\mathrm{sat}}(T) \text{ from Antoine.} \tag{17}$$

These are *experimental* γ_i — the test now asks whether they obey thermodynamics.

8.2 Step 2 — the Gibbs–Duhem point test

The Gibbs–Duhem equation is the constraint every real mixture must satisfy at constant T, P:

$$\sum_i x_i\,\mathrm{d}\ln\gamma_i = 0 \quad \Longrightarrow \quad x_1\frac{\mathrm{d}\ln\gamma_1}{\mathrm{d}x_1} + x_2\frac{\mathrm{d}\ln\gamma_2}{\mathrm{d}x_1} = 0 \ \ \text{(binary)}. \tag{18}$$

The operation evaluates the left-hand side at each interior point (central differences in x_1) and reports its max and mean magnitude — the *point residual*. A consistent set keeps it near zero. (Isobaric data adds a small $-H^{\mathrm{E}}/RT^2 \cdot \mathrm{d}T/\mathrm{d}x$ term, dropped here; stated as a caveat, not hidden.)

8.3 Step 3 — the Herington area test

Integrate (18) across the whole composition range: for an isothermal set Gibbs–Duhem forces the net area of $\ln(\gamma_1/\gamma_2)$ to vanish. Herington's isobaric form compares that net area to the absolute area and corrects for the boiling-range:

$$D = 100\,\frac{\left|\int_0^1 \ln\frac{\gamma_1}{\gamma_2}\,\mathrm{d}x_1\right|}{\int_0^1 \left|\ln\frac{\gamma_1}{\gamma_2}\right|\mathrm{d}x_1}, \qquad J = 150\,\frac{T_{\max} - T_{\min}}{T_{\min}}, \tag{19}$$

(both integrals by the trapezoid rule over the sorted points). The set **passes** when $|D - J| < 10$. The GUI's consistency card plots $\ln\gamma_1$, $\ln\gamma_2$ and $\ln(\gamma_1/\gamma_2)$ vs x_1 with the net area shaded — you SEE the $\pm$ areas cancel, not just the number.

Finally the infinite-dilution coefficients $\gamma_1^\infty, \gamma_2^\infty$ (the dilute-end limits) are reported — the most stringent point of any model later fitted to the set.

> **Runnable case.** `tutorials/props/compare/compare_vle_etoh_water` — the `vleConsistency` op on the Carey–Lewis ethanol/water data (Herington $|D - J| = 4.6 < 10$, PASS).

9 The bench beyond thermodynamics

Two operations ship in the props bench whose subject is not a property package at all: a catalyst pellet and a recorded step response. They are here because the bench already has the right shape — a model, a declared input, a published answer with its own verification beside it.

9.1 One pellet: `thielePellet`

`thielePellet` resolves a `kind catalyst;` asset record and its effective diffusivity, solves the isothermal first-order intraparticle problem numerically, and *verifies* the answer against the closed form — a run missing its declared `verification` tolerances exits 1. It publishes the field $c(r)/c_s$, the Thiele modulus ϕ with the effectiveness factor η it produces, the Weisz–Prater observable $\eta\phi^2$ and the swept $\eta(\phi)$ family over three geometries. The physics lives in `src/unitOperations/reactor/pellet/PelletDiffusion`, where the four reactors that today assume $\eta = 1$ will reach it. Witness `thiele01_sphere_first_order`: CO in a 3 mm alumina sphere at 573.15 K with a teaching $k = 40\ \mathrm{s}^{-1}$ gives $\phi = 1.016$, $\eta = 0.6657$, verified to 1.3×10^{-6}; `thiele02_prater_temperature` adds Prater's temperature field ($\Delta T_{\mathrm{centre}} = 15.57$ K). It never decides whether diffusion matters in the reader's case: it announces its scope on every run — first order, one temperature, one pellet size, both external films neglected — and judges nothing.

9.2 A recorded experiment: `reactionCurve`

Open-loop identification off a trajectory somebody else recorded. `reactionCurve` reads a `choupoCtrl` trajectory, a `manipulated` column and a `measured` one, cuts it into step segments and fits each to the first-order-plus-dead-time surrogate

$$y(t) = y_0 + K\,\Delta u \left[1 - e^{-(t - t_{\mathrm{step}} - \theta)/\tau}\right], \qquad t \geq t_{\mathrm{step}} + \theta,$$

by Nelder–Mead, then applies the named rules (`zieglerNichols`, `cohenCoon`, `imcLambda`). It solves no plant: a missing trajectory refuses. It publishes (K, τ, θ) with the fit's residual, plus segment, curve and tuning CSVs. A segment that has not settled under the declared `settling` test yields no time constant; a dead time within the sampling interval is unresolved, and every rule dividing by θ is refused by name (IMC never does). Witness `tutorials/ctrl/ctrl17_step_reaction_curve` (a `ctrl` case, run first): the composition loop takes all three rules; the temperature loop refuses Ziegler–Nichols and Cohen–Coon. The surrogate is a surrogate: the op never judges its adequacy, simulates the tuned loop, or recommends a rule.

10 Operations reference: what the bench can be asked

Everything `choupoProps` does is one of the operations below, declared in `system/propsDict`:

```
operations
(
    {
        name   myFirstOp;         // yours; names the output
        type   propertyScan1D;    // one of the types below
        ...                       // the type's own keys
    }
);
```

Operations run in the order written, each independent of the others. In the tables, * marks a key the operation requires; everything else has a default or is optional. Each entry names a runnable case, which is the fastest way to see the keys in a real dict.

10.1 Evaluating a state

`propertyPoint` Single-point property evaluation: at a given (T, P, composition) the thermo package reports every property the chosen models can compute (Psat, gamma_i, K_i, Cp_ig, H, S, Z, v, ...). Results print to the run log.

`properties`	Properties to evaluate
`state*`	Thermodynamic state

> **Runnable case.** `tutorials/props/compare/acetone01_ipa_water_azeotrope` (and 8 more)

`explainProperty` The derivation LEDGER of one mixture property at one state: every term with the record field or model it came from (the `s_298/dHf_298` datum, the Cp integral as a difference of two engine calls, the mixing and pressure lines, the EoS residual), re-added and checked against the engine's own assembled value at round-off. The op computes no thermodynamics of its own and REFUSES if its explanation and the engine disagree, if the property is not one it can explain, or if a `pureFluids {}` fundamental-equation override answers the state (that route's ledger is the release's own equation, not datum-plus-integral).

`property*`	Property to explain
`state*`	Thermodynamic state

> **Runnable case.** `tutorials/props/molecular/entropy01_air_ledger`

`exergy` PHYSICAL (thermo-mechanical) flow exergy of a mixture state against a DECLARED dead state: b = (h − h0) − T0·(s − s0), every term the engine's own H_real/S_real at the same composition, both legs published so the total re-adds. The dead state is the problem's declaration — a missing `deadState {}` REFUSES (an exergy is a statement about a state AND an environment, and the engine does not choose the plant's environment). CHEMICAL exergy refuses by name: it needs a standard-environment model (Szargut) nobody has curated.

`deadState*`	Dead state (the environment)
`state*`	Thermodynamic state

> **Runnable case.** `tutorials/props/molecular/exergy01_air_dead_state`

`mixingRules` The van der Waals ONE-FLUID DECOMPOSITION of a cubic EoS's mixture parameters at one state: every pure a_i(T) and b_i, every binary k_ij as constructed, and every pair's additive term in a_mix = Sum_i Sum_j y_i y_j (1 - k_ij) sqrt(a_i a_j), b_mix = Sum_i y_i b_i — plus the identity that re-adds the terms to the a_mix the engine runs (a gap REFUSES). The numbers come from the SAME buildMix the hot path executes; a ternary needs three binary decisions and the table shows which were declared and which run the announced k_ij = 0 default. A model with no one-fluid rule (idealGas, PC-SAFT) refuses by name.

`state*`	Thermodynamic state

> **Runnable case.** `tutorials/props/molecular/mixrules01_natural_gas`

`propertyScan1D` One-dimensional property sweep: varies one state variable (T, P or a mole fraction) over n evenly-spaced points and evaluates the requested properties at each. Writes one CSV row per point.

`output*`	Output file
`properties*`	Properties to evaluate
`state`	Held state
`vary*`	Scan axis

> **Runnable case.** `tutorials/props/compare/acetone01_ipa_water_azeotrope` (and 12 more)

`propertyScan2D` Two-dimensional property sweep: varies two state variables on a regular grid and evaluates each property at every node. Writes one CSV row per node; failed evaluations are written as NaN.

`output*`	Output file
`properties*`	Properties to evaluate
`state`	Held state
`varyX*`	X axis
`varyY*`	Y axis

> **Runnable case.** `tutorials/props/scan/scan2d01_co2_compressibility`

`propertyScanBinary` Binary liquid-liquid map at one (T, P): samples the mixing Gibbs energy g_mix across the whole composition axis AND resolves the binodal by an LL flash of a declared feed. The two views teach each other — the g_mix curve SHOWS the common tangent, the flash FINDS it. The feed must lie INSIDE the miscibility gap or the flash rightly reports one phase; for an asymmetric binodal that is not the equimolar point, which is exactly the lesson the water/1-butanol case carries.

`feed`	LL-flash feed
`grid`	Grid resolution
`output`	CSV output
`state*`	Held state

> **Runnable case.** `tutorials/props/scan/binary01_lle_water_nbutanol`

`propertyScanTernary` Ternary map at one (T, P): sweeps the composition simplex and reports the phase behaviour — the LLE solubility envelope with tie-lines, or a VLE/VLLE audit — from the declared activity model. The map is the model's fingerprint: change the declared model and the envelope moves.

`grid`	Grid resolution
`mode`	Map mode
`output`	CSV output
`shard`	Simplex shard
`state*`	Held state
`tieStride`	Tie-line stride [-]

> **Runnable case.** `tutorials/props/scan/ternary01_audit_vlle` (and 2 more)

`purePhaseDiagram` Pure-component P-T phase diagram: the vapour-pressure curve from the component's own model, and — when the case declares the solid's data — the sublimation and melting curves meeting it at the triple point. The solid block is CASE data (axiom 4): triple point, heats of fusion and sublimation, and the fusion volume change (negative for water, which is why its melting line leans backwards — the lesson the reference case exists for).

`grid`	Grid resolution
`output`	CSV output
`solid`	Solid-phase data

> **Runnable case.** `tutorials/props/scan/phase01_water_full`

`psychrometricChart` Psychrometric chart for ANY carrier + condensable pair, generated from the package's own models rather than copied from a handbook: saturation line, constant-relative-humidity curves and wet-bulb lines over a declared temperature span. Carrier and condensable are NOT interchangeable — only the condensable needs a vapour-pressure model, and the op refuses by name when it has none. Air-water is one instance, not the definition; the reference case runs N2-water.

`P`	Chart pressure [Pa]
`TmaxC`	Upper temperature [-]
`TminC`	Lower temperature [-]
`carrier*`	Carrier (dry) gas
`condensable*`	Condensable component
`grid`	Grid resolution
`output`	CSV output
`relativeHumidity`	RH curves
`wetBulb`	Wet-bulb lines

> **Runnable case.** `tutorials/props/heat/psychro01_n2_water` (and 1 more)

`steamTables` IAPWS-IF97 industrial water/steam properties. This is a propsDict operation under choupoProps (like propertyPoint), NOT a flowsheet unit. At startup it self-checks the transcribed coefficients against the release verification tables and REFUSES the run on any deviation above 1e-8. Give exactly ONE mode block per operation: point, saturation or isobar. CSV columns are canonical SI (h, u, s, cp, cv in J/kg-based units). Tutorial tutorials/props/steam/steam01_if97_verification locks every release verification table as a golden KPI at reltol 1e-8.

`isobar`	Isobaric scan
`output`	CSV output
`point`	Single (p,T) state
`saturation`	Saturation-line scan

> **Runnable case.** `tutorials/props/steam/steam01_if97_verification`

10.2 Mixtures and activity

`activityCoefficients` Liquid activity coefficients of a MOLECULAR mixture at one temperature and composition, straight from the declared activity model. The glass-box counterpart of a VLE calculation: the gammas are what make a flash a flash, and this op prints them on their own so the model's behaviour can be read rather than inferred from a split. Change the declared model and the numbers move — which is the lesson.

`composition*`	Liquid mole composition
`output`	CSV output
`temperature`	Temperature [K]

> **Runnable case.** `tutorials/props/molecular/nrtl01_ethanol_water_package`

`gibbsMap` Equilibrium map over a (T, P) grid by Gibbs minimisation: at every grid point the declared species set equilibrates over the element balance, and the declared metric (a species' mole fraction, an element's recovery) is contoured. The chart's pedagogical weapons are the two DECLARED boxes: an `industrialWindow` (where industry actually operates — for ammonia, NOT where equilibrium is best, because the catalyst rules) and a `kineticBand` (the user-declared region where kinetics kill the approach), both watermarked with their labels so the map says why the plant is not at the thermodynamic optimum.

`Pgrid*`	Pressure grid
`Tgrid*`	Temperature grid
`anchors`	Published anchor points
`elements*`	Elements
`feed*`	Feed moles
`industrialWindow`	Labelled (T, P) box
`kineticBand`	Labelled (T, P) box
`metric*`	Contoured metric
`output`	CSV output
`species*`	Species set
`temperatureApproach`	Equilibrium approach [K]

> **Runnable case.** `tutorials/props/gibbs/nh3_equilibrium_map`

`vleConsistency` Gibbs-Duhem thermodynamic-consistency test on MEASURED binary VLE data. Back-computes the activity coefficients from the data itself by modified Raoult (gamma_i = y_i P / (x_i Psat_i(T))), then reports the pointwise Gibbs-Duhem residual, the Herington area test (pass if |D-J| < 10) and the infinite-dilution gammas. Needs measured y: a bubble-T-vs-x set alone cannot run it, and testing an NRTL or Wilson CURVE is a tautology since those are consistent by construction. The CSV carries x1, lnGamma1, lnGamma2, lnRatio, gdResidual.

`P`	Pressure [Pa]
`component*`	Reference component
`dataset*`	Measured dataset
`output`	CSV output
`partner`	Binary partner
`state`	Held state

> **Runnable case.** `tutorials/props/compare/compare_vle_etoh_water`

`hConsistency` Enthalpy-surface identity audit: checks that the assembled package's enthalpy route is internally consistent for the named components at one temperature. A pure numerical identity on the package's own H surface — it needs no experimental data and answers a question the tutorials cannot: whether the rungs a component's record pins actually compose.

`components*`	Components to audit
`output`	CSV output
`temperature`	Audit temperature [K]

> **Runnable case.** `tutorials/props/thermoTest/h_surface_identity`

10.3 Electrolytes

`speciate` Aqueous speciation: distributes the declared analytical totals over the curated equilibrium network and reports every species, its activity coefficient, the ionic strength, the water activity and the mineral saturation indices. By default it is SI-ONLY — it says which solids are supersaturated without precipitating any. Adding an `equilibrate { minerals ( ... ) }` block lets the NAMED solids precipitate to their SI = 0 ceiling; naming the allowed set is the author's decision and is required, because which solid a brine is permitted to drop is chemistry, not a default.

`activityModel`	Aqueous activity model
`analyticalTotals`	Analytical totals
`atmosphere`	Open-system gas partial pressures
`composition`	Composition form
`diagMeanIonic`	Cation/anion pairs to publish a mean ionic activity coefficient for
`diagSpecies`	Species to pin as diagnostics
`equilibrate`	Allowed precipitation
`network`	Admitted chemistry records
`networkScope`	Speciation network scope
`output`	CSV output
`pH*`	Given pH [-]
`reason`	Restriction reason
`temperature`	Temperature [K]
`totals`	Analytical totals (synonym)
`totalsBasis`	What the totals are
`vapour`	Vapour side (Edwards Eq 5/6/11)
`verifyGlobal`	Run the full catalogue self-check

> **Runnable case.** `tutorials/props/electrolyte/edwards01_sour_water_activity` (and 11 more)

`pitzerActivity` Mean-ionic activity coefficient and osmotic coefficient of ONE salt by the Pitzer ion-interaction model, as a function of molality. The salt is named by its two ions and its stoichiometry is DERIVED from electroneutrality (Ca + Cl gives $CaCl_2$), so the case never restates what the charges already say. The pair parameters come from the curated catalogue; a missing pair is an error, not a zero.

`anion*`	Anion species
`cation*`	Cation species
`molality`	Molality scan
`output`	CSV output
`temperature`	Temperature [K]
`validation`	Reference points

Runnable case. `tutorials/props/electrolyte/archer01_nacl_cold_to_hot` (and 6 more)

`electrolyteActivity` Mean-ionic activity and osmotic coefficients of the salt the CASE's declared thermophysical system carries — the package's own electrolyte surface, exercised directly. Where `pitzerActivity` names its salt by two ions and is the model's bench, this op takes the salt from the declaration and is the PACKAGE's bench: the numbers here are exactly what the flowsheet units will see, which is the point of running it before them.

`molality`	Molality scan
`output`	CSV output
`temperature`	Temperature [K]

Runnable case. `tutorials/props/electrolyte/pitzer02_nacl_package`

`enrtlMultiSalt` Multi-salt symmetric eNRTL bench: the mean-ionic gammas of two (or more) salts sharing an ion, swept across the mixing fraction at constant total molality — the Harned-rule experiment. Two formulations run side by side by declaration: `published` (the industrial expressions exactly as printed) and `refined` (gamma as the EXACT Gibbs-Duhem-consistent derivative of the same excess Gibbs energy, the chain rule kept through the composition-dependent mixing rules). The difference between them is measured, not asserted.

`formulation`	Gamma formulation
`output`	CSV output
`salts*`	Salts
`sweepPoints`	Sweep points [-]
`tauEE`	Ion-ion tau [-]
`totalMolality`	Total molality [mol/kg]

Runnable case. `tutorials/props/electrolyte/enrtl_multisalt_nacl_kcl`

`enrtlMixedSolvent` Mixed-solvent eNRTL bench: the mean-ionic activity coefficient of a salt in water-alcohol mixtures, regressed system by system against published data. The case SPELLS OUT the binary tau parameters and the alcohol's own constants so it reads as its own citation — defaults exist (the published water-ethanol-NaCl set) but a case that states them teaches where every number came from. Each entry of `systems` is one solvent composition with its Debye-Hückel A_phi and its measured (m, gamma) points; the op reports the deviation per system.

`alcohol`	Alcohol constants
`output`	CSV output
`salt`	Salt tau parameters
`systems*`	Solvent compositions with data

Runnable case. `tutorials/props/electrolyte/enrtl_mixed_nacl_ethanol_esteso`

`scalingScan` Concentration scan of a water analysis, the RO/evaporator scaling audit: the declared analysis is concentrated point by point along a recovery axis, the full speciation is solved at each point, and every mineral's saturation index is reported — WHERE each scale becomes possible, before any membrane is sized. Takes the same solution-definition keys as `speciate` (totals, pH incl. `solve`, atmosphere for an open system, temperature, equilibrate for allowed precipitation); adds the recovery axis and, with an equilibrate block, a `feedFlow` that turns precipitation into kg/day. `feedFlow` without `equilibrate` is REFUSED — it only feeds a rate an SI-only scan does not compute — and a bare number is refused too: it needs a volumetric unit.

`activityModel`	Aqueous activity model
`analyticalTotals`	Water analysis
`atmosphere`	Open-system gas pins
`diagSpecies`	Species to pin as diagnostics
`equilibrate`	Allowed precipitation
`feedFlow`	Feed flow [m3/s]
`output`	CSV output
`pH`	pH
`recovery`	Recovery axis
`temperature`	Temperature [K]
`totals`	Water analysis (synonym)

Runnable case. `tutorials/props/electrolyte/pitzer_vs_davies_ro_concentrate` (and 4 more)

exchange Ion-exchange equilibrium on the props bench: the declared water contacts a resin at a declared dose and the op reports the exchanged composition, the resin loading and the pH. The `exchange {}` block accepts resin, resinDose and CEC and NOTHING else — the capacity nameplate lives in the resin's own record, and an unknown key refuses by name with the grammar. `pH solve;` asks for the proton balance to be solved rather than given.

`analyticalTotals`	Analytical totals
`diagSpecies`	Species to pin as diagnostics
`equilibrate`	Allowed precipitation
`exchange*`	Resin contact
`output`	CSV output
`pH`	pH
`temperature`	Temperature [K]
`totals`	Analytical totals (synonym)

> **Runnable case.** `tutorials/props/electrolyte/softener_brackish`

10.4 Fitting and estimating

fitParameters Levenberg-Marquardt regression of thermophysical-model parameters against experimental data, and the SCORING of a package as it stands. Two modes, and the difference matters: `fit` (default) adjusts the listed parameters and writes a fit log (chi^2 and the parameter trajectory per iteration) plus a parity CSV at the optimum; `evaluate` changes nothing and only scores — 'how good are the pairs I already have against MY data?' — so in that mode a `parameters` list is OPTIONAL and, when present, is a pinned what-if whose entries carry `value` rather than the fit vocabulary of initial/min/max.

`acceptance`	
`dataset`	
`evidence`	
`mode`	Mode
`options`	Solver options
`output`	Output files
`parameters`	Parameters to vary
`residual*`	What is being fitted

> **Runnable case.** `tutorials/props/adsorption/isotherm02_fit_jaschik` (and 7 more)

vaporPressureFit Least-squares fit of measured vapour-pressure data to the Antoine form, producing the A, B and C a component record carries. Like `heatCapacityFit`, this is curation made visible: the residuals are reported beside the coefficients so the fit is judged, not just taken.

`acceptance`	
`component`	Component
`dataset`	Measured dataset
`evidence`	
`output`	CSV output

> **Runnable case.** `tutorials/props/curation/curate01_water_evidence_partition` (and 1 more)

heatCapacityFit Least-squares polynomial fit of measured heat-capacity data, producing the `coefficients ( ... )` list a component record carries. The fit is a CURATION act made visible: the residuals and the recovered coefficients are reported so the student can judge the fit before promoting it into a `.dat`.

`acceptance`	
`component`	Component
`dataset`	Measured dataset
`degree`	Polynomial degree [-]
`evidence`	
`output`	CSV output

> **Runnable case.** `tutorials/props/fit/cp01_polynomial_recovery`

estimateComponent Group-contribution estimation of a component's constants — curation made visible, like the fit ops: the estimate is computed, compared against a reference where one is given, and written as a promotable draft, never silently into the catalogue. THREE routes by `model`: a group method (Joback for small molecules, vanKrevelen for a polymer repeat unit) decomposes the molecule into `groups`; a scalar-anchored method (RiaziDaubert) builds a petroleum PSEUDO-component from `anchors { Tb; SG; }` and needs no groups — the record it writes says loudly that a cut is a LUMP of hundreds of species, not a molecule; and the polymer route adds a `polymer {}` block whose packing factor k is visible, never hidden, because rho = M0/(k·Vw) and k IS the assumption.

`anchors`	Anchor data
`component`	Component name
`derived`	Derived-property requests
`groups`	Structural groups
`model`	Estimation model
`output`	CSV output
`polymer`	Polymer options
`reference`	Reference values
`validation`	Reference values (synonym)

> **Runnable case.** `tutorials/plant/polycaprolactonePlant/properties` (and 6 more)

10.5 Benches

`kinetics1D` One-dimensional batch kinetics bench: integrates c(t) for an nth-order rate, overlays a measured dataset when one is declared, and — with a `fit` list — regresses the rate constant or the Arrhenius pair (k0, Ea) from the data. The MODEL-DISCRIMINATION lesson lives here: the same dataset run at order 1 and order 2 shows which residuals carry structure, which is adequacy, not consistency.

`component`	Component
`dataset`	Measured dataset
`fit`	Parameters to fit
`output*`	CSV output
`rate*`	Rate law
`state`	State block
`time`	Time span

> **Runnable case.** `tutorials/props/kinetics/rate01_arrhenius_recovery`

`heatTransferBench` Self-check of the heat-transfer correlation library against its own textbook worked examples: every built-in correlation is evaluated at its published example's conditions and held to the printed answer within a declared tolerance. Runs with NO configuration — the bench IS the test — and the tolerances exist only to honour textbook rounding and chart-reading scatter, never to loosen a failing correlation.

`aadTolerance`	Deviation tolerance [-]
`aadTolerancePhaseChange`	Phase-change tolerance [-]

> **Runnable case.** `tutorials/props/heat/htbench01_correlations`

`isothermEval` Evaluates a curated adsorption isotherm — the loading of one adsorbate on one adsorbent as a function of pressure and temperature — from the adsorbent record's own parameters. The glass-box surface of what a fixed-bed or TSA unit uses internally, so the isotherm a breakthrough case runs on can be plotted and checked before the bed is ever solved.

`adsorbate*`	Adsorbate
`adsorbent*`	Adsorbent
`grid*`	Evaluation grid
`output`	CSV output
`unitTwin`	Unit to mirror

> **Runnable case.** `tutorials/props/adsorption/isotherm01_eval_13x_co2`

`elementalComposition` Glass-box surface of the ONE shared formula parser (src/thermo/ElementComposition) that every balance diagnostic uses. Decomposes chemical formulas into their per-element atom counts and reports them, so the parser a student relies on for the element balance can be exercised and read directly. Give `formulas`, `components`, or both; a formula it cannot parse is REFUSED by name rather than silently skipped.

`components`	Catalogue components
`formulas`	Literal formulas
`output`	CSV output

> **Runnable case.** `tutorials/props/thermoTest/atoms01_formula_parser`

10.6 Other

`enthalpyConcentration` The saturated-liquid and saturated-vapour enthalpy loci of a BINARY at ONE pressure — the data an enthalpy-concentration (Ponchon-Savarit) construction is drawn on; the construction itself belongs to a drawing tool. Every row is one equilibrium: it carries BOTH ends of a tie line (the liquid x1 with h_liquid, the vapour y1 with H_vapour) at the ONE temperature they share, so a consumer never pairs rows by index and never interpolates one end of a tie. `role bubble` rows put x1 on the uniform grid, `role dew` rows put y1 on it; both are solved through the SAME bubble-point kernel, a dew node being the inversion y_eq(x) = y1. The op REFUSES up front unless the thermophysical system has EXACTLY 2 components, and unless every component can be priced on the project's one enthalpy datum (the elements at 298.15 K): it names any component with no route to that datum, any component with no idealGasHeatCapacity{} block (no saturated-vapour leg), and any component whose standardThermochemistry.referenceState is off the ideal-gas rung — no private Cp integral is ever substituted. A node that cannot be solved (bubble Newton not converged, a non-monotonic y_eq(x1) that has no single-valued dew inversion, a y1 outside the converged bubble image, an enthalpy kernel refusal) is PUBLISHED as a refusal row with its status, never dropped. Diagnostics headline lambda = H_vapour − h_liquid over the locus, the spread of which measures what constant molar overflow costs.

`grid`	Composition grid
`output*`	CSV output
`state*`	The pressure of the diagram

> **Runnable case.** `tutorials/props/scan/hxy01_ethanol_water_1atm`

`freezingPoint` Freezing-point curve T_f(m) of a single salt in water, solved as chemical-potential equality between an actual crystallising SolidPhase (ice) and the Pitzer surface's water activity: f_solid(T) = a_w(m,T)*Psat(T). The first case-reachable consumer of the crystal; its announced approximations (no dCp term, sub-freezing Psat extrapolation) fire on every run. Diagnostics: T_f and depression at the grid end, the dilute-limit slope against nu*K_f (the derived cryoscopic constant), and the depression AAD against a measured dataset via the standard validation block.

`anion*`	Anion species
`cation*`	Cation species
`molality`	Molality scan
`output`	CSV output
`temperature`	Temperature [K]
`validation`	Reference points

> **Runnable case.** `tutorials/props/electrolyte/fpd01_nacl_freezing`

`frictionBench` The Darcy friction factor's correlation library, in two parts. VERIFY (always): every registered correlation — Blasius, Colebrook-White, Haaland, Churchill — reproduces its own published anchor, and the deviation is printed beside the correlation's validity window and its primary citation. COMPARE (optional): all four are evaluated at the SAME Re and roughness, and the spread is published TWICE — over all of them, and over only those inside their stated windows. The two are reported apart because collapsing them would say 'the correlations disagree by 26 %' when the truth is 'three agree within 2 % and one was asked a question it was never fitted for'. The bench never says which correlation is right: that depends on the pipe, and ranking them would hide the choice the engineer has to make.

`compare`	Comparison point
`deviationTolerance`	Anchor deviation tolerance [-]

> **Runnable case.** `tutorials/props/hydraulics/moody01_friction_correlations`

`phaseEnvelope` The P-T phase envelope of a MIXTURE at fixed feed composition: the bubble curve (V/F -> 0) and the dew curve (V/F -> 1) traced in the (T, P) plane and written as a long-form CSV (P_Pa, T_K, curve). An envelope is drawn to find the CRICONDENTHERM – the highest temperature at which the feed can hold a liquid, what a pipeline is designed against – and the retrograde region beside it, and both live at the nose, which no single-specification march can close: this trace does NOT close it, says so on every run, and names Michelsen's arc-length continuation with a switching specification (Fluid Phase Equilib. 4 (1980) 1) as the algorithm that would. The bubble and dew solves are BubblePoint::compute and DewPoint::compute – the same routines the bubbleT and dewT units call, so neither saturation residual exists twice. Marching in pressure, the temperature maximum is not a turning point for the specification, so an INTERIOR maximum is reported as a BRACKETED cricondentherm naming the two pressures it lies between; an endpoint maximum is not reported, because it says where the range stops and not where the mixture peaks. A point that CONVERGED but is discontinuous with its branch is discarded and the branch ends there: the bubble curve is monotonically increasing in P up to the critical point (physics, not a heuristic) and the dew curve gets a continuity test, because both saturation solvers bracket T in 200-700 K and are warm-started, so Newton can land on a different root and report success – which it did, drawing a bubble temperature FALLING with pressure and publishing a cricondentherm that was the resulting spike. A branch that produces no point at all is reported as never having started, which is a different finding from a branch that stopped, and has a different remedy.

`composition*`	Feed composition
`output`	CSV output
`pressure*`	Pressure march

> **Runnable case.** `tutorials/props/molecular/envelope01_natural_gas` (and 1 more)

`reactionCurve` OPEN-LOOP IDENTIFICATION off a trajectory somebody else recorded. Given a CSV written by a choupoCtrl run, one column driven and one column watched, it cuts the record into step SEGMENTS, fits a first-order-plus-dead-time surrogate (K, tau, theta) to each by Nelder-Mead on the sum of squared residuals, and applies the classical tuning rules to the segment the reader nominates. It solves no plant and integrates nothing: the experiment must already exist on disk, and the op reads it. THE SURROGATE IS A SURROGATE — the record is whatever the plant did, an FOPDT is three numbers, and the statement that the second is standing in for the first reaches the reader at the site AND in the end-of-run caveat block. A segment whose response has not SETTLED yields no time constant, because both K and tau are read off the end of the curve; the steady-state test that decides this is declared, not assumed. A dead time smaller than the record's own sampling interval is UNRESOLVED, and every rule that divides by theta (Ziegler-Nichols, Cohen-Coon) is then REFUSED BY NAME for that segment — a controller gain computed from a number the experiment could not measure is a number with nothing behind it. IMC survives there, because its algebra never divides by theta. NOTHING IN CHOUPO EVER CLOSES A LOOP ON THESE SETTINGS: the published gains are what each rule says, not a claim that they work.

`manipulated*`	The driven column
`measured*`	The watched column
`minimumPoints`	Shortest usable segment [-]
`output`	CSV output
`reference`	Which segment the headline quotes
`settling`	The steady-state test
`trajectory`	The recorded experiment
`tuning`	Tuning rules to apply

> **Runnable case.** `tutorials/ctrl/ctrl17_step_reaction_curve`

`solubilityParameter` The Hildebrand solubility parameter, delta = sqrt((dHvap(T) - R T) / V_m), for every component of the case, plus the pairwise |delta_i - delta_j| table that is what the question usually is: the smaller the gap, the more likely two liquids mix. Delta is DERIVED and never stored – the latent heat comes from the same Watson correlation the enthalpy legs use, and the molar volume from the record's Vliq – so a stored value would be a second home for a fact the record already fixes. Where a record declares a published delta as an ANCHOR, the deviation is printed beside the derived number; that column is a check on the derivation and never an input to it, and it exists because all three ways this arithmetic fails are silent (a dropped RT term, a wrong V_m unit, and Pa^0.5 reported where the literature uses MPa^0.5 – a factor of 31.6 that still reads as a solubility parameter). Refuses by name for a component with no Tc, no HvapTb or no Vliq, and at a temperature at or above Tc, where there is no liquid and delta is not defined. States on every call that Hildebrand carries no hydrogen bonding – it cannot separate a polar solvent from a hydrogen-bonding one of the same cohesive energy density, so a close pair is a CANDIDATE and never a verdict – and names Hansen's three-parameter split as what would, not implemented because its parameters are a dataset this project does not hold.

`T`	Temperature [K]
`output`	CSV output

> **Runnable case.** `tutorials/props/molecular/solubility01_hildebrand_ladder`

`thielePellet` The PROPS BENCH for one catalyst pellet: it resolves the pellet from its asset record and its effective diffusivity from the ONE home that owns that resolution, solves the isothermal first-order intraparticle boundary-value problem numerically, and VERIFIES that answer against the closed form. It publishes the concentration field c(r)/c_s, the Thiele modulus phi with the effectiveness factor eta it produces, the Weisz-Prater observable modulus eta*phi^2, and — swept — the classical eta(phi) family over the three geometries. The physics is NOT here: it lives in unitOperations/reactor/pellet/PelletDiffusion, which the four reactors that today assume eta = 1 will reach. The op judges nothing: it publishes phi, eta and the field, announces what it assumed (isothermal, first-order, one pellet size, external film neglected), and leaves the reader to decide whether diffusion matters in their case. A run whose numerical answer misses the declared verification tolerances exits 1.

`P`	Pressure [Pa]
`T*`	Temperature [K]
`catalyst*`	Catalyst pellet
`diffusion*`	Diffusion route
`nodes`	Grid points [-]
`output`	CSV output
`rateConstant`	Rate constant, per pellet volume [1/s]
`rateConstantMass`	Rate constant, per catalyst mass [m3/(kg.s)]
`refinement`	Grid-refinement study
`species*`	Diffusing species
`sweep`	eta(phi) family
`thermal`	Temperature field (Prater)
`verification`	Verification tolerances

> **Runnable case.** `tutorials/props/reactor/thiele01_sphere_first_order` (and 1 more)

This guide grows with the props pillar. Planned: a second group-contribution method (Constantinou–Gani) to compare against Joback; group contribution for liquid viscosity; a structure (SMILES) front-end for the compound bench.